

Multi-Armed Bandits for Adaptive Retrieval

DSA4213 Final Project

Group 41: Zhang Zhenjie

National University of Singapore

e0957630@u.nus.edu

Outline

- 1 Introduction
- 2 Multi-Armed Bandits
- 3 Methodology
- 4 Results
- 5 Conclusion
- 6 References

Motivation: The One-Size-Fits-All Problem

Standard RAG pipelines use fixed retrieval strategies for all queries.

Why this is inefficient:

- **Simple queries** ("What is the capital of France?")
→ Complex retrieval is wasteful
- **Complex queries** ("Who is the wife of the director of the movie that won Best Picture in 2020?")
→ Simple retrieval fails

Our Goal: Dynamically adapt retrieval strategy based on query complexity, balancing **accuracy** and **computational cost**. (Tang et al. 2025)

Solution: Multi-Armed Bandit Approach

Key Idea: Treat retrieval strategy selection as a reinforcement learning problem

- Each retrieval method (BM25, dense vector, hybrid) is an **arm**
- For each query, the system:
 - 1 Selects an arm probabilistically
 - 2 Receives a reward based on accuracy and latency
 - 3 Updates its policy to improve future selections
- Learns optimal strategy through **exploration vs. exploitation**

MAB: Core Concepts



Objective: Maximize cumulative reward by balancing exploration and exploitation

The Exploration-Exploitation Tradeoff:

- **Exploitation:** Choose the currently best-performing arm
- **Exploration:** Try other arms to discover better options

Estimation of Reward & Update

Estimated Reward (Sample Average):

$$Q_t(a) = \frac{1}{N_t(a)} \sum_{i=1}^{t-1} R_i \cdot \mathbb{I}_{A_i=a}$$

- $N_t(a)$: Number of times action a chosen by time t
- R_i : Reward received at time i

Incremental Update:

$$Q_{t+1}(a) = Q_t(a) + \frac{1}{N_t(a)} (R_t - Q_t(a))$$

- Estimate is updated by a step size (which decreases with the number of pulls) in the direction of the error (the difference between the received reward and the current estimate).

Dynamic Reward Function

Our reward balances accuracy and computational cost:

$$R_t = A(y, \hat{y}_a) - \lambda \cdot C_{norm}(a)$$

- $A(y, \hat{y}_a)$: Accuracy metrics (e.g. Exact Match, F1-score)
- $C_{norm}(a)$: Normalized latency cost of arm a
- λ : Cost-performance trade-off parameter
- Higher $\lambda \rightarrow$ stronger penalty for slow retrieval

Epsilon-Greedy Algorithm

Strategy: Balance exploration and exploitation with probability ϵ

Algorithm 1 select_arm()

- 1: Generate a random number $p \sim \text{Uniform}(0, 1)$
 - 2: **if** $p < \epsilon$ **then** ▷ Explore
 - 3: Return a random arm $a_t \in \{1, \dots, k\}$
 - 4: **else** ▷ Exploit
 - 5: Return the arm with the highest average reward:
 - 6: $a_t = \arg \max_a Q[a]$
 - 7: **end if**
-

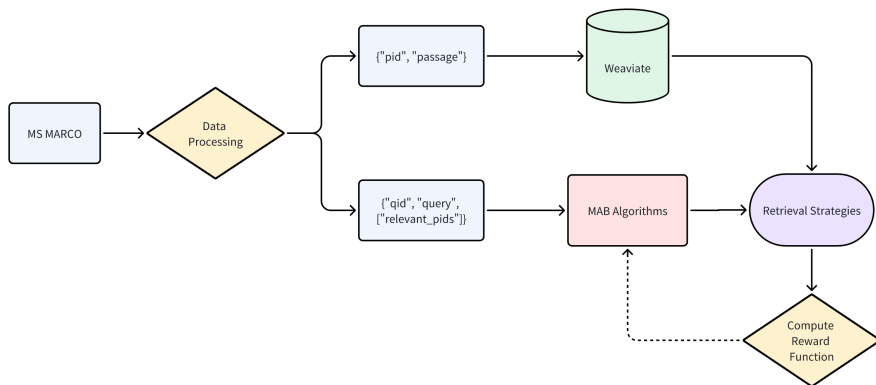
Algorithm 2 update(arm_index, reward)

- 1: Let $a \leftarrow \text{arm_index}$ and $R_t \leftarrow \text{reward}$
 - 2: $N[a] \leftarrow N[a] + 1$ ▷ Increment the pull count
 - 3: $Q[a] \leftarrow Q[a] + \frac{1}{N[a]}(R_t - Q[a])$ ▷ Update Q-value
-

Focus: Adaptive retrieval (not full RAG pipeline)

- **Dataset:** MS MARCO (queries, passages, relevance mappings)
- **MAB Algorithms:** Epsilon-greedy, UCB, Thompson Sampling
- **Retrieval Methods (Arms):**
 - BM25 Search (fast, keyword-based)
 - Dense Vector Search (slower, semantic)
 - Hybrid Search (combination)
- **Database:** Weaviate (via Docker)

Project Flow



Initial Results

Method	Avg Latency (ms)	Avg Accuracy
BM25 Search	1.76	0.682
Dense Vector Search*	7.31	0.903
Hybrid Search ($\alpha = 0.5$)	13.25	0.901
Epsilon-Greedy ($\epsilon = 0.1$)	7.16	0.896
UCB ($c = 2$)	7.95	0.893
Thompson Sampling	7.76	0.902

*Dense encoder: multi-qa-MiniLM-L6-cos-v1

Key Finding: MAB algorithms achieve near-optimal accuracy at half the latency of hybrid search

Learning Dynamics

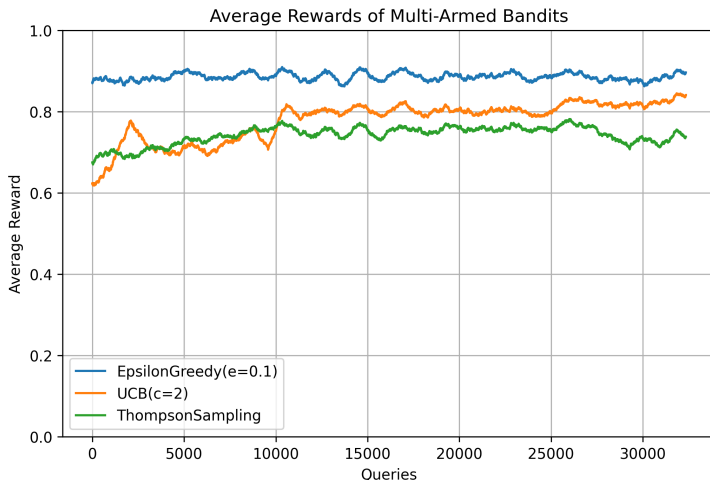


Figure: All algorithms converge and learn to select effective retrieval methods

Key Takeaways & Future Work

Contributions:

- Demonstrated **MAB algorithms** can adaptively select retrieval strategies
- Achieved strong **accuracy-latency balance** without heuristics
- All bandit methods learn effective policies over time

Future Directions:

- **Contextual Bandits:** Use query embeddings to guide arm selection
- **Full RAG Pipeline:** Extend to joint retrieval-generation optimization
- **Richer Rewards:** Incorporate answer quality, diversity, and user feedback
- **Cross-dataset Validation:** Test on Natural Questions, TriviaQA

References I

-  Tang, Xiaqiang et al. (2025). *MBA-RAG: a Bandit Approach for Adaptive Retrieval-Augmented Generation through Question Complexity*. arXiv: 2412.01572 [cs.AI]. URL: <https://arxiv.org/abs/2412.01572>.
-  Lewis, Patrick et al. (2021). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*. arXiv: 2005.11401 [cs.CL]. URL: <https://arxiv.org/abs/2005.11401>.
-  Jeong, Soyeong et al. (2024). *Adaptive-RAG: Learning to Adapt Retrieval-Augmented Large Language Models through Question Complexity*. arXiv: 2403.14403 [cs.CL]. URL: <https://arxiv.org/abs/2403.14403>.
-  Bajaj, Payal et al. (2016). "Ms marco: A human generated machine reading comprehension dataset". In: *arXiv preprint arXiv:1611.09268*.

Thank You